VIT Vidyalankar Institute of Technology
Accredited A+ by NAAC

Version 09/27-09

Consolidated Academic Administration Plan for the Course

E-waste & Environmental management (core as per NBA)
Sem. V – Program INFT 2026-27 Odd Semester
Faculty - Prof. Sonaali Borkar (Cluster Mentor), Prof. Nilima Main

The academic resources available in VIT –

VMIS (ERP)	V-Refer and V-Live	VIT Library	VAC & MOOC Courses
Institute & Department Vision and Mission	Former IA question papers and solutions (prepared by faculty)	Former IA question papers solutions - hardcopy	Value Added Courses (VAC) are conducted throughout the semester & in the semester break - Enrol for the VACs
Program Educational Objectives (PEO)	MU end semester examination question papers and solutions (prepared by faculty)	MU end semester exam question paper & solutions - by faculty, hardcopy	
Program Specific Outcome (PSO)	Class notes and Digital Content for the subject (scanned / typed by faculty)	All text books, reference books, e -books mentioned in the syllabus & AAP	Online courses from NPTEL, Coursera etc. are pursued throughout the semester - Register for the course & get certified
Program Outcome (PO)	Comprehensive question bank, EQ, GQ, PPT, Class Test papers	Technical journals and magazines for reference	
Departmental Knowledge Map	Academic Administration Plan & Beyond Syllabus Activity report	VIT library has many resources e:g :- IEEE, Nimbus, xplore, EBSCO etc.	Watch former lectures captured in LMS at VIT

1.a Course Objectives (Write in detail – as per NBA guidelines)

Cognitive	What do you want students to know?	To enable the students to apply the knowledge of engineering materials to design technically sound system and adapt to new technologies.
Affective	What do you want students to think / care about?	To make the students understand human, social, ethical and environmental context of their profession and contribute positively to the needs of individuals and society.
Behavioural	What do you want students to be able to do?	To inculcate research and development ability and enable the students to analyse real life problems.

Advice to Students:

Attend every class!!! Missing even one class can have a substantial effect on your ability to understand the course. Be prepared to think and concentrate, in the class and outside. I will try to make the class very interactive. Participate in the class discussions. Ask questions when you don't understand something. Keep up with the class readings. Start assignments and homework early. Meet me in office hour to discuss ideas, solutions or to check if, what you understand is correct.

The v-Refer Link http://vidyalankarlive.com/vrefer/index.php/f/1086833

Creation of microsite or teams link – (Division wise)

TE INFT: General | TE INFT- EWEM- NSM

Collaboration Policy:

We encourage discussion between students regarding the course material. However, no discussion of any sort is allowed with anyone on the assignment and homework for the class. If you find solution to some problems in a book or on the internet, you may use their idea for the solution; provided you acknowledge the source

(name and page in the book or the website, if the idea is found on the internet). Even though you are allowed to use ideas from another source, you must write the solution in your own words. If you are unsure whether or not certain kinds of collaboration is possible, please ask the teacher.

1.b Course Outcome (CO) Statements and Module-Wise Mapping (follow NBA guideline)

CO No.	Statements	Related Module/s
CO1	Understand need and concept of e-waste management & recycling	Module 1
CO2	To gain the knowledge of legislative rules and regulations related to E-waste management.	Module 4
CO3	Recognize the tools and technologies required for e-waste recycling assessment and monitoring.	Module 2
CO4	Address contemporary issues and analyse the local and global impact of computing and engineering solutions on individuals, organizations, and society	Module 5, 6
CO5	Apply the knowledge about E-waste management in routine daily life to minimize the hazards.	Module 3

1.c Course Outcome (CO) Statements mapping with Learning Outcome (LO)

	Mapped to Learning Outcomes
CO1	1.1, 1.2, 1.3, 1.4
CO2	4.1, 4.2, 4.3
CO3	2.1, 2.2, 2.3
CO4	5.1, 5.2, 5.3, 6.1, 6.2, 6.3
CO5	3.1, 3.2

1.d Mapping of COs with POs (As per new detailed Syllabus)

(List of POs is available in V-refer)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
	Knowledge A	Analysis	Design	Investigation	Modern Tools	Society	Environment &sustainability	Ethics T	Teamwork C	Communication	Project Mgt	Lifelong learning
CO 1	S	M	M	M	W	M	S	-	S	-	-	S
CO 2	S	M	M	M	W	M	S	-	S	-	-	S
CO 3	S	M	M	M	W	M	S	-	S	-	-	S
CO 4	S	M	M	M	W	M	S	-	S	-	-	S

CO 5	S	M	M	M	W	M	S	-	S	-	-	S

1.e Mapping of COs with PSOs (As per new detailed Syllabus)

	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	W	S	S	NA
CO 2	W	S	S	NA
CO 3	W	S	S	NA
CO 4	W	S	S	NA
CO 5	W	S	S	NA

1.f Teaching and Examination Scheme (As specified by the autonomous syllabus) for the Course

Verticals	BSC/ESC	Program Courses	Multidisciplinary Courses	Skill Courses	HSSM	Experiential Learning	Liberal Learning
Tick suitable category					√		

Subject Code	Subject Name	**Teaching Scheme**			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/Practical	Tutorial	Total
VEC02	E-waste & Environmental management	2	-		2	-	--	2

Subject Code	Subject Name				**Examination Scheme**				
		Theory			**Total (Theory)**	**Practical**			**Total (Practical)**
		ISA	MSE	ESE		ISA	ESE	ORAL	
VEC02	E-waste & Environmental management	75	--	--	75	--	--	--	--

Subject Code	Subject Name	**MSE-1***		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
NA				

Subject Code	Subject Name	**MSE-2***		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
NA				

Subject Code	Subject Name	ESE#		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
NA				

*** Recommended distribution: -** 30 Marks from Assignments, 40 marks based on assignments with slightly enhance difficulty /complex, 30 marks from thought provoking

Recommended distribution: - 30 Marks from Assignments, 40 marks based on assignments/MSE with slightly enhance difficulty /complex, 30 marks from thought provoking

1.g Faculty-Wise Distribution of all Lecture-Practical-Tutorial Hours for the Course

Divisions	Lecture (Hrs.)	Practical (Hrs.)				Tutorial (Hrs.)			
		Batch 1	Batch 2	Batch 3	Batch 4	Batch 1	Batch 2	Batch 3	Batch 4
TE INFT (90 students)	2- NSM	--	--	--	--	--	--	--	--

1.h Office Hours (Faculty will be available in office in this duration for solving students' query)

Division	Day	Time (at least 1 Hr. / Division)	Venue (Office Room No.)
TE INFT (90 students)	Thursday	4.00 to 5.00 pm	D101, faculty room

Note: Office hours should be after academic hours of students.

2.a Syllabus: Module Wise Teaching Hours and % Weightage in autonomous syllabus Question Paper

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
1	Overview of E-waste & its management: Introduction, toxicity due to hazardous substances in e-waste and their impacts, domestic e-waste disposal, e-waste management, technologies for recovery of resource from electronic waste, guidelines for environmentally sound management of e-waste, occupational and environmental health perspectives of recycling e-waste in India.	06	100%	--	--	
Learning Outcome-1.1	Explain the concept of e-waste, identify hazardous substances in electronic waste, and analyze their impacts on human health and the environment.					
Learning Outcome-1.2	Assess domestic e-waste disposal practices and explore effective strategies for sustainable e-waste management.					
Learning Outcome-1.3	Describe technologies for resource recovery from e-waste and summarize guidelines for the environmentally sound management of electronic waste.					

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
Learning Outcome-1.4	Analyze the occupational and environmental health challenges associated with e-waste recycling in India and propose measures to mitigate these risks.					
2	E-Waste Recycling: Technologies for recovery of resources from electronic waste, resource recovery potential of e-waste, steps in recycling and recovery of materials-mechanical processing, technologies for recovery of materials. Recovery of materials from e-waste, recovery of metals from e-waste	05	100%	--	--	
Learning Outcome-2.1	Describe the technologies used for the recovery of resources and materials from e-waste, focusing on mechanical processing and advanced recovery methods.					
Learning Outcome-2.2	Assess the resource recovery potential of e-waste, including valuable metals and other materials, and understand its significance for sustainable resource management.					
Learning Outcome-2.3	Explain the step-by-step process of e-waste recycling and material recovery, highlighting the role of different technologies in efficient resource extraction.					
3	Exposure pathway of pollutants emitted from Recycling of E-Waste: Environmental and public health issues E-waste health risk assessment	04	100%	--	--	
Learning Outcome-3.1	Explain the exposure pathways of pollutants emitted during e-waste recycling and their effects on environmental and public health.					
Learning Outcome-3.2	Conduct a health risk assessment related to e-waste recycling, identifying key risks and proposing strategies to minimize adverse health impacts.					
4	E-Waste Management Rules of India (2011 and 2016 Rules): Regulations in India - Hazardous Waste (Management, Handling and Transboundary Movement) Rules, 2008 - E-waste (Management and Handling) Rules, 2011; and E-Waste (Management) Rules, 2016 - Salient Features and its likely implication. Government assistance for TSDFs. E-waste Management: Case Studies and Unique Initiatives from around the World The international legislation: The Basel Convention; The Bamako Convention. The Rotterdam Convention. Waste Electrical and Electronic Equipment (WEEE) Directive in the European Union, Restrictions of Hazardous Substances (RoHS) Directive	06	100%	--	--	
Learning Outcome-4.1	Explain the evolution and salient features of Indian e-waste management rules (2011 and 2016), the Hazardous Waste Rules (2008), and the government's assistance for Treatment, Storage, and Disposal Facilities (TSDFs).					
Learning Outcome-4.2	Compare international e-waste management legislation, including the Basel, Bamako, and Rotterdam Conventions, and directives such as WEEE and RoHS, highlighting their implications for sustainable e-waste practices.					

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
Learning Outcome-4.3	Assess case studies and unique e-waste management initiatives from around the world, identifying best practices and their potential applicability in different contexts.					
5	Technological Advances to overcome Environmental problems: • Concept of Green Buildings, Various indoor air pollutants and their effects on health • Carbon Credit: Introduction and general concept. • Disaster Management: Techniques of Disaster • Management to cope up with (i) Earthquake and (ii) Flood. Remote sensing and GIS – Introduction and its applications in environment sector	04	100%	--	--	
Learning Outcome-5.1	Explain the concept of green buildings and identify various indoor air pollutants along with their health impacts.					
Learning Outcome-5.2	Describe the concept of carbon credit and evaluate disaster management techniques to mitigate the impacts of earthquakes and floods.					
Learning Outcome-5.3	Understand the basics of remote sensing and GIS and analyze their applications in addressing environmental challenges.					
6	Environmental Assessment and Management: Environment Impact Assessment (EIA), Introduction to various symbols Environmental audit and Eco-labelling, ISO - 14001, 18001 and 31001 Pollution Control Legislation- Functions and powers of Central and State Pollution Control Board. Environmental Clearance, Consent and Authorization Mechanism.	05	100%	--	--	
Learning Outcome-6.1	Explain the concept and significance of Environmental Impact Assessment (EIA), environmental audits, eco-labeling, and relevant ISO standards (14001, 18001, and 31001).					
Learning Outcome-6.2	Describe the functions and powers of the Central and State Pollution Control Boards in enforcing pollution control legislation.					
Learning Outcome-6.3	Understand the processes involved in obtaining environmental clearance, consent, and authorization, and their role in ensuring compliance with environmental regulations.					
	*** Insert rows for more modules in the Course** **Total**	**30**	**100**			

Note: - As an attachment Annexure is required for assessment criteria of learning outcomes.

2.b Prerequisite Courses

No.	Semester	Name of the Course	Topic/s
1	12th	Environmental science	--
2			

3			

2.c Relevance to Future Courses

No.	Semester	Name of the Course
1	7 or 8	Environmental Management
2		
3		

2.d See :- Identify real life scenarios/examples which uses the knowledge of the subject ,(Discussion on how to prepare examples and case studies e.g. "Boeing Plane": C Programming Language – Intro to Computer Science – Harvard's CS50 (2018) – Bing video)

Real Life Scenario	Concept Used
Disposal of used, not functioning electronic gadgets like Computers, televisions, mobile phones, refrigerators, charging cables, etc.	E-waste disposal & recycling
E-waste management rules in India	Hazardous Waste (Management, Handling and Transboundary Movement) Rules, 2008 E-waste (Management and Handling) Rules, 2011

3 Past Results – Division-Wise

Details	Target – DEC 2026	MAY 2026	DEC 2025	MAY 2025
Course Passing % – Average of 2 Divisions	100%	NA	100%	97.86%
Marks Obtained by Course Topper (mark/100)	75/75	NA	74/75	73/75

	INFT A		Division B		Division C	
Year	Initials of Teacher	% Result	Initials of Teacher	% Result	Initials of Teacher	% Result
Dec 2025	NSM	100%	NSM	100%	NSM	100%
May 2025	NSM	97.86%	NSM	100%	NSM	100%
DEC 2022	NSM	100%	NSM	100%	NSM	100%

4 All the Learning Resources – Books and E-Resources

4.a List of Textbooks (T – Symbol for Textbooks) to be Referred by Students

Sr. No	Textbook Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library

1	Environmental Science	G. Tyler Miller Jr. and Scott Spoolman	Brooks/Cole	13	--	Yes
2						
3						
4						
5						

4.b List of Reference Books (R – Symbol for Reference Books) to be Referred by Students

Sr. No	Reference Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Electronic Waste Management	Hester R.E., and Harrison R.M.	Science	1	7, 8, 24, 25,	No
2	Electronic Waste	Fowler B.	(Toxicology and Public Health Issues), Elsevier	1	13, 14	No
3	E-waste: implications, regulations, and management in India and current global best practices	Johri R.	TERI Press, New Delhi	1	1, 3, 5	No
4	Environmental Management, Text Book and Cases	Krishnamoorthy B.	PHI Learning (P) Ltd., New Delhi	1	2, 3, 6, 7, 8, 9	No
	Environmental Science and Engineering	Keerthinarayana and Daniel Yesudian	Hi-Tech Publications, 2008	1	1 to 4	Yes

4.c List of E - Books (E – Symbol for E-Books) to be Referred by Students

Sr. No	E- Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Electronic Waste Management Rules 2016	--	Central Pollution Control Board, Delhi	1	--	No
2	Waste electrical and electronic equipment (WEEE): Flows, quantities and management, a global scenario	Mihai F.C., Gnoni M.G., Meidiana C., Ezeah, C. , Elia V	HAL Open Science	--	--	No

4.d Reading latest / top rated research papers (at least 5 papers)

Name of Paper		Published in	Problem Statement	Available in our Library

	Name of Authors (Background)	Date	Journal		
E-Waste: Current Research and Future Perspective on Developing Countries	L. Halim and Y. Suharyanti	June 2020	International Journal of Industrial Engineering and Engineering Management (IJIEEM)	Concerns regarding e-waste because of increasing rate of e-waste production and its hazardous content	No
E-Waste Management	Saravanamurthy. D, Raja Raja Chozhan. V. E, Sarangan. B, Bharathkumar. R	2018	International Journal of Engineering Research & Technology (IJERT)	This investigation intends to exhibit that the present worldwide e- waste situations and well-being perils could drag out the accomplishment of manageable improvement targets.	No
E-waste Management for Environmental Sustainability: an Exploratory Study	Rahul S. Mor, Kuldeep Singh Sangwan, Sarbjit Singh, Atul Singh, Manjeet Kharub	2021	Science Direct	This paper aims to reveal the role of e-waste management practices for environmental sustainability and explores people's awareness level about e-waste management, its generation, and primary treatment practices in educational institutions.	No
E-Waste Management: As a Challenge to Public Health in India	Monica and Jugal Kishore	2010	Indian Journal of Community Medicine	Disposal of e-waste is an emerging global environmental and public health issue, as this waste has become the most rapidly growing segment of the formal municipal waste stream in the world. Therefore, the need of proper e-waste management has been realized.	No
E-Waste Management in India: A Study of Current Scenario	Dr. Neha Garg, Deepak Kumar Adhana	2019	International Journal of Management, Technology and Engineering	The present paper highlights the scenario of e-waste in India and other parts of globe. It also exhibits the trends of e-waste in India through a comparison with other countries.	No

4.e **Based on research paper an identify the current Problem statement**

Problem Statement			Used in			
	Quiz	Assignment	*Mini Project*	Poster Presentation	Test	Any Other
Finding environment-friendly recovery technique which can promote e-waste as resources for production			√			Presentation
Analysing the flow of EEE products sold in developing countries is a challenge due to smuggling, unrecorded local production, and product traceability						Flow chart diagram
Survey study: Carry out a study of e-waste awareness programs amongst metro city and a small town people.						Online Survey
Many workers are involved in crude dismantling of these electronic items for their livelihood and their health is at risk; therefore, there is an urgent need to plan a preventive strategy in relation to health hazards of e-waste handling among these workers in India.	√					Case study
As a consumer, how to dispose your e-waste?					√	

4.f **Identify Companies / Industries which use the knowledge of the subject and thus may provide Internships and final Placements**

Name of the Company	To be / Contacted for		
	Student Internship	Student Final Placement	Faculty Internship
Eco Recycling Ltd (Ecoreco), Mumbai	Yes		May be

4.g **Identify suitable relevant TOP Guest Speakers from Industry, Example: - (CS50 Lecture by Mark Zuckerberg - 7 December 2005 - YouTube)**

Name of the Identified Guest Speaker	Designation	Name of the Company
Dr. Vaitea Cowan	Co-founder	Enapter, Green Hydrogen Berlin Germany
Dr. Cathy Mulzer	Research Investigator	DuPont, Columbus, Ohio Metropolitan Area

4.h **Identify relevant technical competitions to participate [Competitions -Paper Presentations, Projects, Hackathons, IVs etc..]**

Name of the Relevant Technical Competition Identified to participate	Organized by	Date of the Event
Technical competitions by ACM, ITSA	INFT dept.	To be declared soon
SIH	Govt. of India	To be declared soon

4.i **Identify faculty in TOP schools / Universities who are teaching same / similar subject and develop rapport e.g. Exchange Lecture Material (Assignments / Tests / Project etc..), Joint Paper Publication**

University	Name of the Course	Name of Faculty	Type of Collaboration		
			Exchange of Lecture Material	Joint Publication/ Research	Other
IIT KHARGPUR	Electronic Waste Management – Issues And Challenges	Prof. Brajesh Kumar Dubey	√		

4.j **Module Best Available in – Title of the best resource [from *4.a* to *4.d* in this AAP] and other details as necessary**

Module No.	Title of the Module	Textbook	Mention the Tile					
			Reference Book	E-books	Journal	E-Journal	Available in our Library	V-refer
1	Overview of E-waste & its management		1				No	Yes
2	E-Waste Recycling		2				No	Yes
3	Exposure pathway of pollutants emitted from Recycling of E-Waste		1				No	Yes

4	E-Waste Management Rules of India		2				No	Yes
5	Technological Advances to overcome Environmental problems		1				No	Yes
6	Environmental Assessment and Management		2				No	Yes
7								

4.k **Referred to any top-rated university in that subject for content**

University	Name of the Course	Name of Faculty	Date of Delivery of the Course	Remarks
IIT Kharagpur	Electronic Waste Management - Issues and Challenges	Prof. Brajesh Kumar Dubey	--	Course might be available in the new academic semester, old course content is available on NPTEL website

4.l **Faculty received any certification related to this subject. List of Certifications Identified / Done**

Course	Certifying Agency	No. of Hours	Level of the Course		Certification		Remarks
			Introductory	Advance Skill Development	Done on	Proposed to be on	
Electronic Waste Management - Issues And Challenges	NPTEL	30 hrs		√		October 2026	Electronic Waste Management - Issues And Challenges

4.m **Completed subject wise/cluster wise training with cluster mentor. List of relevant Refresher Course Identified / Done**

Course	Certifying Agency (As suggested by DAB/Cluster Mentor/Industry/University other than MU)	Certification		Remarks
		Done on	Proposed to be on	
Pedagogy	The Complete Storytelling Course	June 2025		
PBL				
Sub. Content Training	Udemy Learn the Basics of Waste Management:		October 2026	

	Manage & Reduce Waste			

4.n Best Practices Identified and adopted

No.	Item	Best Practices Identified		
		IIT Kharagpur	Harvard	IIT Bombay
1	Microsite			
2	Video Lectures	√		
3	Assignments	√		√
4	Mini Project			
5	Assessment Metric			
6	Quizzes	√		
7	Labs/ Practical (PBL)			
8	Tests			
9	Peer Assessment	√		
10	Any Other			

4.o Web Links for Online Notes/YouTube/VIT Digital Content/VIT Lecture Capture/NPTEL Videos

Students can view lectures by VIT professors, captured through LMS 'Lecture Capture' in VIT campus for previous years.

No.	Websites / Links	Module Nos.
1	https://youtu.be/CmP67zq5hfo Digital India : e-Waste Management	1
2	https://youtu.be/wsNtsH7jHz0 Perspective: Electronic Waste Management | 13 December, 2021	1
3	https://youtu.be/4kG3HTAo6xI E-Waste: Definition, sources, classification, collection, segregation, treatment and disposal	1
4	https://youtu.be/MQLadfsvfLo What is E-WASTE Pollution? | What Causes Electronic Waste?	2
5	https://youtu.be/4JYtUKujZfY Eco India: How a start-up in New Delhi has been processing 70% of India's formal e-waste	3
6	https://youtu.be/ntEVHTqsq6A Dangers of e-waste | e-waste management rules,2016	4
7	https://youtu.be/QRsqHKIpNss Lecture 53: E-Waste Management	2
8	https://youtu.be/S2lmPla1iWE How 6 Million Pounds Of Electronic Waste Gets Recycled A Month	3
9	https://youtu.be/9WIwlljva_s	5

	Disaster Management | Disasters - Types and Effects	
10	https://youtu.be/RWtT0EfhNsE Environmental impact assessments: identifying relevant issues and concerns	6

4.p Recommended MOOC Courses like Coursera / NPTEL / MIT-OCW / edX/VAC etc.

Sr. No.	MOOC Course Link	Course conducted by – Person / University / Institute / Industry	Course Duration	Certificate (Y / N)
1	Waste Management 4.0 - The Waste Mgmt Industry in Industry 4	Udemy		Y

5 Consolidated Course Lesson Plan

	From (date/month/year)	From (date/month/year)	Total Number of Weeks
Semester Duration	06/07/2026	23/10/2026	15

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
1	1	All	Introduction to EWEM course & explaining AAP **Private study 0**		All	L1, L2		
	2					L1. L2		
2	3	1	Overview of E-waste & its management: Introduction, toxicity due to hazardous substances in e-waste and their impacts,		CO1	L1, L2, L3		
	4	1	domestic e-waste disposal		CO1	L3		
3	5	1	e-waste management, technologies for recovery of resource from electronic waste		CO1	L2		
	6	1	guidelines for environmentally sound management of e-waste,		CO1	L2		
4	7	1	occupational and environmental health perspectives of recycling e-waste in India.		CO1	L2		
	8	2	E-Waste Recycling: Technologies for recovery of resources from electronic waste,		CO1	L2		
5	9	2	resource recovery potential of e-waste,		CO1	L2 & L3		
	10	2	steps in recycling and recovery of materials-mechanical processing		CO1	L2		

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
6	11	2	technologies for recovery of materials		CO1	L2		
	12	2	Recovery of materials from e-waste, recovery of metals from e-waste		CO1	L2 & L3		
7	13	3	Exposure pathway of pollutants emitted from		CO5	L2 & L3		
	14	3	Recycling of E-Waste: Environmental issues		CO5	L2		
8	15	3	Public health issues		CO5	L2 & L3		
	16	3	E-waste health risk assessment		CO5	L2 & L3		
9	17	4	E-Waste Management Rules of India (2011 and 2016 Rules): Regulations in Indi		CO2	L2		
	18	4	- Hazardous Waste (Management, Handling and Transboundary Movement) Rules, 2008		CO2	L2		
10	19	4	- E-waste (Management and Handling) Rules, 2011; and E-Waste (Management) Rules, 2016 - Salient Features and its likely implication. Government assistance for TSDFs.		CO2	L2		
	20	4	E-waste Management: Case Studies and Unique Initiatives from around the World		CO2	L2		
11	21	4	The international legislation: The Basel Convention; The Bamako Convention. The Rotterdam Convention.		CO2	L2		
	22	4	Waste Electrical and Electronic Equipment (WEEE) Directive in the European Union, Restrictions of Hazardous Substances (RoHS) Directive		CO2	L2		
12	23	5	Technological Advances to overcome Environmental problems: Concept of Green Buildings, Various indoor air pollutants and their effects on health		CO4	L2		
	24	5	Carbon Credit: Introduction and general concept.		CO4	L2		

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
			Disaster Management: Techniques of Disaster					
13	25	5	Remote sensing and GIS – Introduction and its applications in environment sector		CO4	L2		
	26	6	Environmental Assessment and Management: Environment Impact Assessment (EIA)		CO3	L2		
14	27	6	Introduction to various symbols Environmental audit and Eco-labelling,		CO3	L2		
	28	6	ISO - 14001, 18001 and 31001 Pollution Control Legislation-		CO3	L2		
15	29	6	Functions and powers of Central and State Pollution Control Board.		CO3	L2		
	30	6	Environmental Clearance, Consent and Authorization Mechanism.		CO3	L2		

6 Rubric for Grading and Marking of Term Work (inform students at the beginning of semester)

- Activity/ies should be designed as per reference of credit structure.
- If the subject is of 2 credit, activity/ assignment should be design for 2 hours with appropriate complexity and engaging time.

Theory (ISA= 75 M)											Total
Class Participation	Online game quiz	video task	Carbon Footprint Calculator	Device Anatomy	Class test	Case study	Data Analysis Activity	Sustainability Reel / Podcast	Green Hackathon	Campus Sustainability Consultancy (E-Waste Audit + Awareness Campaign)	
--	05	05	05	05	10	10	10	10	10	10	75

Practical (ISA= NA)	Total

Class Participation	Practical -1	Practical -2	Practical -3	Practical -4	Practical -5	Practical -6	Practical -7	Practical -8	Practical -9	Practical -10	Practical -11	

Class Participation	MSE-1	MSE-2	ESE	Total
75	--	--	--	75

7 Assignments / Tutorials Details

Assignment/ Tutorial No.	Title of the Assignments / Tutorials	CO Map	Mapping Bloom Taxonomy Level	Assignment/ Tutorials given to Students on	Assignments to be submitted back on
0	Assignment zero	All	L2	Week 1	Week 2
1	**Activities** Online Game based competition (tool- Kahoot) **(5M)**	CO1	L3	Week 3	Week 3
2	Video based task (tool- Edpuzzle) **(5M)**	CO1	L3	Week 4	Week 4
3	Carbon Footprint Calculator (Assessment of personal and digital carbon footprint with sustainability recommendations) **(5M)**	All	L2	Week 5	Week 5
4	Device Anatomy: Exploring Electronics from Inside (Identification of Components, Hazardous Materials, Recyclable Parts and Resource Recovery Opportunities) **(5M)**	All	L2	Week 6 & 7	Week 6 & 7
5	Class test- followed by peer assessment **(10M)**	CO1 & CO3	L3	Week 8	Week 8 & 9
6	Case studies- E-waste management in developed countries & developed countries **(10M)**	CO3	L2	Week 6 & 7	Week 6 & 7
7	Data Analysis Activity: Use datasets related to global e-waste generation to visualize using Python, Excel, or Tableau. **(10M)**	CO1, 3 & 5	L3	Week 8 & 9	Week 8 & 9

8	Sustainability Reel / Podcast **(10M)**	CO4	L3		
9	Green Hackathon- Technology Solutions for E-Waste Management (Mini project) **(10M)**			Week 14 & 15	Week 14 & 15
10	Campus Sustainability Consultancy (E-Waste Audit + Awareness Campaign) **(10M)**	All	L3	Week 10	Week 10

Analysis of Assignment / Tutorial Questions and Related Resources

Assignment / Tutorial No.	Week No.	Type* (√)			Module No.	Based on #			Question Type (√)	
		OT	CS	DTP		Textbook	Reference Book	Other Learning Resource	Real Life Assignments	Thought Provoking
0	1				All			√		
1	2	√			1			√		√
2	3	√			1 & 3			√	√	√
3	5			√	2	√	√		√	√
4	6	√	√	√	1, 2 & 3	√		√		√
5	7		√		All	√		√		√
6	9			√	All			√	√	√
7	10		√		1, 2, 3 & 4			√		√
8	12	√	√		5 & 6	√	√		√	√
9	13		√		All	√		√		√
10	14		√		All	√		√		√

* Tick (√) the Type of the Assignment: Online Tools (OT); Collaborative Assignments (CS); Design /Thought provoking (DTP)
Write number for textbook, reference book, other learning resource from this AAP – *from Points 4.a to 4.d*

8 In Semester Assessment (ISE) / Other Class Test / Open Book Test (OBT)/Take Home Test (THT) Details

Tests	Test Dates	Module No.	CO Map	MSE Question Paper Pattern	Policy
ISA	Week 1 to week 15	1 to 6	CO1 to CO5	--	--
Mobile based quiz	Week 2	1	CO1	--	--
Class test	Week 5	1 & 2	CO1 & CO3	--	--

Crossword puzzle	Week 9	1, 2 & 3	CO1, CO3 & CO5	--	--
Class tests / prelims					
Class tests / prelims					
Any other test/exams					

*** Failures of IA test (IA1+IA2) shall appear for IA test in the next semester. There is no provision for re-test in the same semester.**

9. Practical Activities

Practical No.	Module No.	Title of the **Experiments**	Type of Experiment		Topics to be highlighted	CO Map
			PBL	Newly Added		
1	NA					
2						

10 Uncovering syllabus with different Activities.

No.	Type of the Activity	Activities	Number of beneficiaries	Other Details – guest profile, feedback, mark sheet, report
1	**Experiential learning/Interaction with Outside World**	1- Guest Lectures by Industry Expert		--
		2- Workshops		Data analysis workshop
		3- Mini Project		Mini projects on E-waste management
		4- Industrial Visit		Visit to Ecoreco, Mumbai
		5- Any other activity		Online studies
2	**Collaborative & Group Activity**	6- Poster Presentation		--
		7- Minute Papers		Module 1
		8- Students Seminars		Module 5 & 6
		9- Students Debates		All- day 1 activity in class
		10- Panel Discussion / Mock GD		--

		11- Mock Interview		--
		12- Any other activity		--
3	**Co-Curricular Activity**	13- Informative videos (NPTEL/YouTube /TEDx/ MIT OW/edX)		NPTEL video
		14- Lecture Capture Usage		--
		15- Any other activity		Edpuzzle tool- Video task
4	**Tests & Assessments**	16- Class Tests/ Weekly Tests		Class test- Module 2
		17- Pop Quiz		--
		18- Mobile App Based Quiz		Quiz on- Module 1 & 2
		19- Open Book Test		Module 3 (part of flipped class activity)
		20- Take Home Test		--
		21- Any other activity		Crossword puzzle

11 AAP

No.	Programme	Course	Uploaded on V-refer	Date
1	TE INFT	E-waste management	Yes	18/7/2026

12 Lecture Guide

No.	Programme	Course	Uploaded on V-refer	Date
1	TE INFT	E-waste management	Yes	18/7/2026

*** Do not delete any activity. Give details for planned events. Write 'NA' for activity Not Planned.**

Consolidated Academic Administration Plan Prepared by (mention all theory teaching faculty names with signature)

Please write below your name and sign with date of the external cluster mentor meeting

Prof Nilima Main	Faculty 2	Faculty 3

External Industry Mentor	External Academic Mentor	VIT Cluster Mentor	Program HOD

Annexure:

Assessment Criteria of Learning Outcomes:

Learning Outcomes: The Learner will:	**Assessment Criteria:** The Learner can:	**Evaluated under ISA/MSE/ESE/LAB**
Explain the concept of e-waste, identify hazardous substances in electronic waste, and analyze their impacts on human health and the environment.	- Define e-waste and list examples of electronic waste. - Identify hazardous components (e.g., lead, mercury, cadmium). - Analyze and describe their effects on human organs and ecological systems.	ISA
Assess domestic e-waste disposal practices and explore effective strategies for sustainable e-waste management.	- List common disposal methods used in households. - Evaluate their environmental implications. - Propose alternative sustainable practices (e.g., extended producer responsibility).	ISA
Describe technologies for resource recovery from e-waste and summarize guidelines for the environmentally sound management of electronic waste.	- Explain technologies like hydrometallurgy, pyrometallurgy, and biometallurgy. - Summarize CPCB/UNEP guidelines for safe disposal. - Distinguish between compliant and non-compliant recycling methods.	ISA
Analyze the occupational and environmental health challenges associated with e-waste recycling in India and propose measures to mitigate these risks.	- Identify unorganized sector practices in India. - Assess exposure risks for workers and communities. - Suggest engineering, legal, and educational mitigation strategies.	ISA
Describe the technologies used for the recovery of resources and materials from e-waste, focusing on mechanical processing and advanced recovery methods.	- Describe steps in mechanical shredding, sorting, and refining. - Differentiate mechanical and chemical processes. - Identify recovery rates and efficiency.	ISA
Assess the resource recovery potential of e-waste, including valuable metals and other materials, and understand its significance for sustainable resource management.	- List valuable recoverable materials (e.g., gold, copper, palladium). - Estimate recovery potential in given samples. - Link recovery to circular economy and sustainability.	ISA
Explain the step-by-step process of e-waste recycling and material recovery, highlighting the role of different technologies in efficient resource extraction.	- Create a flowchart of the e-waste recycling process. - Explain roles of different equipment (e.g., crushers, eddy current separators). - Identify loss points and recommend optimizations.	ISA
Explain the exposure pathways of pollutants emitted during e-waste recycling and their effects on environmental and public health.	- Define key pollutants released during informal recycling. - Identify air, soil, and water exposure pathways.	ISA

	- Assess long-term public health risks using case data.	
Conduct a health risk assessment related to e-waste recycling, identifying key risks and proposing strategies to minimize adverse health impacts.	- List exposure pathways and calculate dose estimates. - Identify critical health endpoints (e.g., neurotoxicity, respiratory illness). - Propose PPE usage, medical surveillance, and policy changes.	ISA
Explain the evolution and salient features of Indian e-waste management rules (2011 and 2016), the Hazardous Waste Rules (2008), and the government's assistance for Treatment, Storage, and Disposal Facilities (TSDFs).	- State timeline and milestones in e-waste law development. - Highlight key changes in 2016 rules (e.g., PROs, EPR). - Describe TSDF model and support mechanisms.	ISA
Compare international e-waste management legislation, including the Basel, Bamako, and Rotterdam Conventions, and directives such as WEEE and RoHS, highlighting their implications for sustainable e-waste practices.	- Summarize the scope and objective of each convention. - Create a comparison matrix for compliance requirements. - Analyze India's obligations under Basel Convention.	ISA
Assess case studies and unique e-waste management initiatives from around the world, identifying best practices and their potential applicability in different contexts.	- Identify at least three case studies (e.g., Japan, EU, Rwanda). - Describe the model followed in each. - Evaluate scalability in Indian context.	ISA
Explain the concept of green buildings and identify various indoor air pollutants along with their health impacts.	- Define green building and key parameters (e.g., energy, water use). - List common indoor pollutants (e.g., VOCs, formaldehyde). - Describe associated health symptoms and mitigation.	ISA
Describe the concept of carbon credit and evaluate disaster management techniques to mitigate the impacts of earthquakes and floods.	- Explain carbon credit trading and GHG mitigation. - Describe structural/non-structural disaster preparedness methods. - Propose disaster-resilient tech solutions.	ISA
Understand the basics of remote sensing and GIS and analyze their applications in addressing environmental challenges.	- Define RS and GIS with examples. - Demonstrate mapping of landfill or contaminated sites. - Explain use in flood prediction or forest cover analysis.	ISA
Explain the concept and significance of Environmental Impact Assessment (EIA), environmental audits, eco-labeling, and relevant ISO standards (14001, 18001, and 31001).	- Describe EIA process steps and legal mandate. - List key indicators for eco-audit and ISO certifications. - Identify products that carry eco-labels.	ISA
Describe the functions and powers of the Central and State Pollution Control Boards in enforcing pollution control legislation.	- List roles such as issuing consent, enforcing law, monitoring quality. - Differentiate CPCB and SPCB functions. - Explain how boards ensure compliance.	ISA
Understand the processes involved in obtaining environmental clearance, consent, and authorization, and their	- Describe steps in environmental clearance process. - Fill sample consent forms or mock applications.	ISA

role in ensuring compliance with environmental regulations.	- Justify the importance of each approval in project cycles.	